

Phase 6: Project Testing

Model Evaluation & System Verification

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Model Evaluation

The XGBoost model was rigorously tested during the training phase using 5-Fold Cross-Validation combined with RandomizedSearchCV.

- **Algorithm Used:** XGBoost Classifier
- **Optimization:** RandomizedSearchCV (50 iterations)
- **Validation Metric:** 5-Fold Cross-Validation Accuracy
- **Result:** The model achieved high consistency across all folds, proving it generalizes well to unseen data and avoids overfitting.

System Verification Cases

The deployed Flask application was tested with various inputs to ensure the data pipeline (String → Int → DataFrame → Prediction) functions correctly.

Test Case		Input Sample		Expected Output	Actual Output	Status
Valid Input (Approved)		Gender=1, come=5000, Credit_Hist=1	In-	APPROVED	APPROVED	PASS
Valid Input (Rejected)		Gender=0, come=1000, Credit_Hist=0	In-	REJECTED	REJECTED	PASS
Boundary Test (Zero Income)		Income=0, CoappIn-		REJECTED	REJECTED	PASS
Invalid Input (Empty Field)		Income field blank	left	Form Validation Error	Form Validation Error	PASS